

Question:

Eliza's rate per hour for the first 40 hours she works each week is \$10. She also receives an overtime pay of 1.2 times her regular hourly rate. If Eliza worked for 45 hours this week, how much are her earnings for this week?

Reasoning + Answer:

Let's break down the problem step by step .

1. **Identify the regular and overtime hours : **

- Eliza works 45 hours this week .
- The first 40 hours are at her regular rate .
- The remaining hours are considered overtime .
- Therefore , the number of overtime hours is $(45 - 40 = 5)$ hours .

2. **Calculate the regular earnings : **

- Eliza's regular hourly rate is \$10 .
- She worked 40 regular hours .
- So , her regular earnings are $(40 \times 10 = 400)$ dollars .

3. **Calculate the overtime earnings : **

- Eliza's overtime pay rate is 1.2 times her regular hourly rate .
- Her regular hourly rate is \$10 , so her overtime rate is $(1.2 \times 10 = 12)$ dollars per hour .
- She worked 5 overtime hours .
- So , her overtime earnings are $(5 \times 12 = 60)$ dollars .

4. **Calculate the total earnings : **

- Her total earnings are the sum of her regular earnings and her overtime earnings .
- So , her total earnings are $(400 + 60 = 460)$ dollars .

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- Eliza worked 45 hours in total .
- The first 40 hours are at her regular rate of \$10 per hour , so her regular earnings are $40 * 10 = \$400$.
- The remaining 5 hours are at her overtime rate , which is 1.2 times her regular rate , so her overtime rate is $10 * 1.2 = \$12$ per hour .
- Her overtime earnings are $5 * 12 = \$60$.
- Her total earnings for the week are $400 + 60 = \$460$.

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<answer> 460 </answer>